

**LAB PROGRAMS (16-20)**  
**COMPUTER VISION-ITA-0505**  
**SLOT-D**

**NAME: DHARSHITHA.K**

**ROLL NO: 192521093**

**QUESTION-16**

Implement a Sobel algorithm using Open CV to filter the input image.

**PYTHON CODE:**

```
import cv2

# Read the image
image = cv2.imread("sample.jpg")

# Check if the image is loaded
if image is None:
    print("Error: Image not found!")
    exit()

# Convert image to grayscale
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Display original image
cv2.imshow("Original Image", image)

# Display grayscale image
cv2.imshow("Gray-scale Image", gray_image)

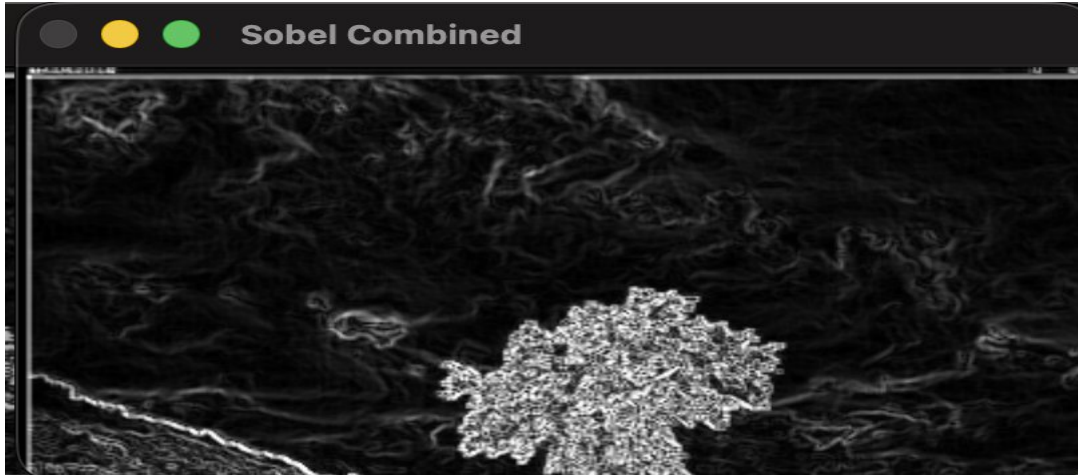
# Wait for a key press
cv2.waitKey(0)
```

```
# Close all windows
```

```
cv2.destroyAllWindows()
```

**OUTPUT:**





### QUESTION-17:

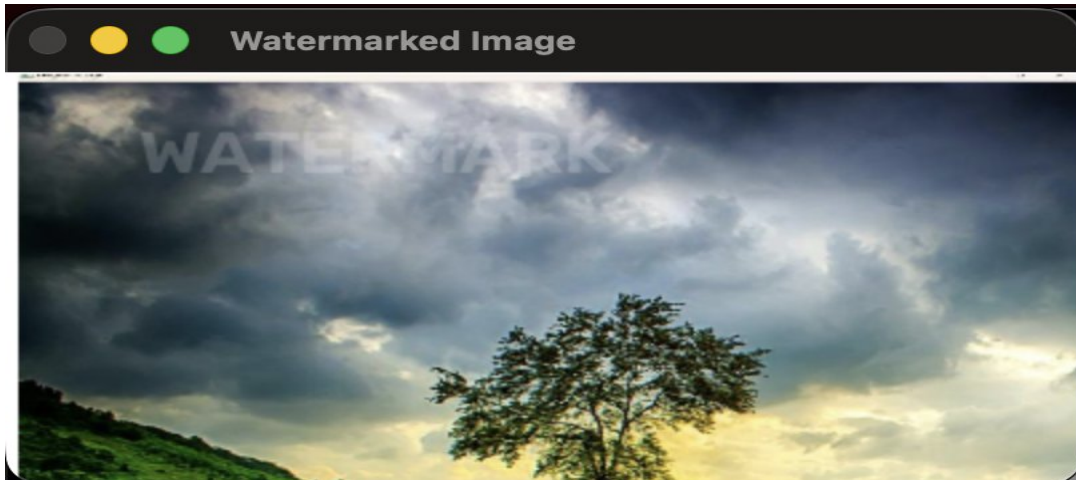
Design and implement a water marking technique to insert the watermark into the original effectively image using Open CV.

### PYTHON CODE:

```
import cv2
# Read the image
image = cv2.imread("sample.jpg")
# Check if the image is loaded
if image is None:
    print("Error: Image not found!")
    exit()
# Apply Gaussian Blur
blur_image = cv2.GaussianBlur(image, (15, 15), 0)
# Display original image
cv2.imshow("Original Image", image)
# Display blurred image
cv2.imshow("Gaussian Blur Image", blur_image)
# Wait until a key is pressed
cv2.waitKey(0)
# Close all windows
```

**cv2.destroyAllWindows()**

**OUTPUT:**



**QUESTION-18:**

implement image cropping, copying and pasting to select a region of interest (ROI) from the source image using Open CV.

**PYTHON CODE:**

```
import cv2

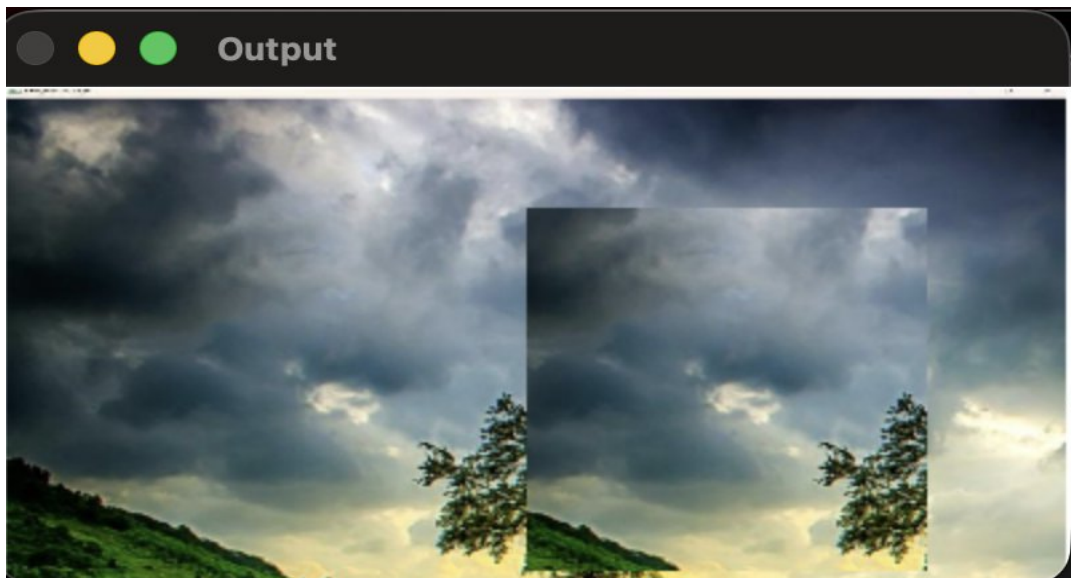
# Read the image
image = cv2.imread("sample.jpg")

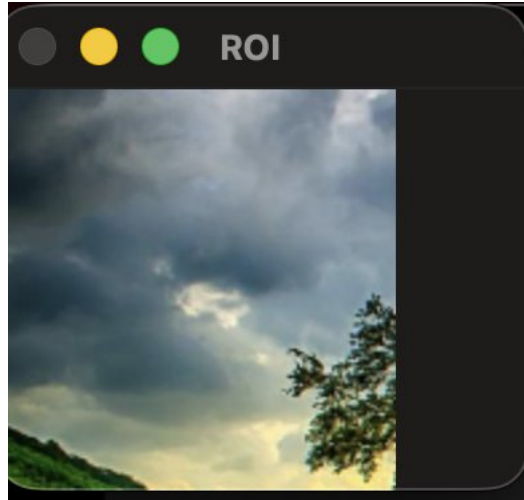
# Check if image is loaded
if image is None:
    print("Error: Image not found!")
    exit()

# Convert to grayscale
```

```
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Apply Canny edge detection
edges = cv2.Canny(gray_image, 100, 200)
# Display images
cv2.imshow("Original Image", image)
cv2.imshow("Edge Detected Image", edges)
# Wait for key press
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

**OUTPUT:**





### **QUESTION-19:**

Implement the Erosion Morphological operations technique using Open CV in python.

### **PYTHON CODE:**

```
import cv2

# Read the image
image = cv2.imread("sample.jpg")

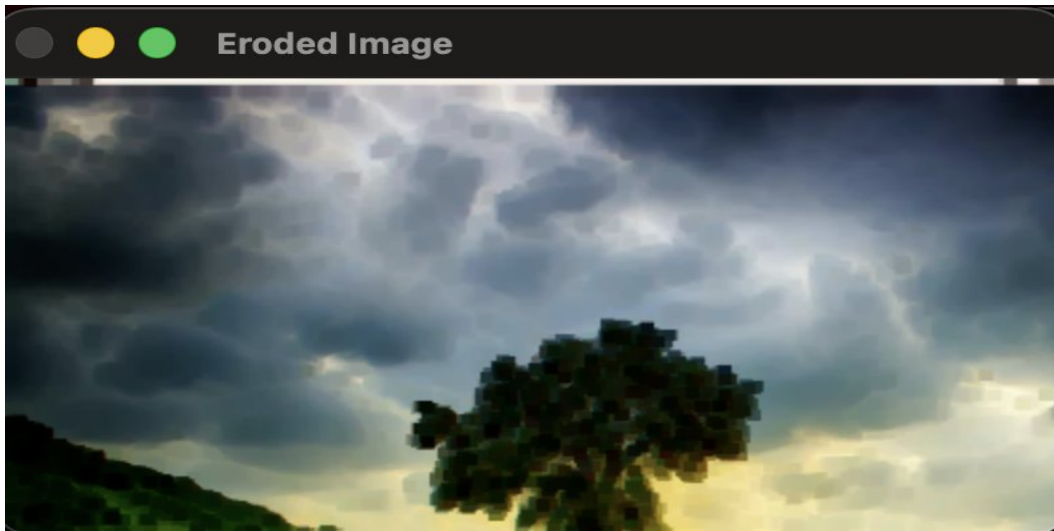
# Check if image is loaded
if image is None:
    print ("Error: Image not found!")
    exit ()

# Convert to grayscale
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Apply Histogram Equalization
```

```
equalized_image = cv2.equalizeHist(gray_image)
# Display images
cv2.imshow("Original Grayscale Image", gray_image)
cv2.imshow("Histogram Equalized Image", equalized_image)
# Wait until a key is pressed
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

**OUTPUT:**



**QUESTION-20:**

Implement the Dilation technique as a Morphological operation to dilate the foreground regions based on Open CV.

**PYTHON CODE:**

```

import cv2
import matplotlib.pyplot as plt
def analyze_histogram(image_path):
    # Read the image
    image = cv2.imread(image_path)
    # Check if image is loaded
    if image is None:
        print("Error: Image not found!")
        return
    # Display the original image
    cv2.imshow("Original Image", image)
    # BGR color channels
    colors = ('b', 'g', 'r')
    # Create histogram plot
    plt.figure(figsize=(8, 5))
    for i, color in enumerate(colors):
        histogram = cv2.calcHist([image], [i], None, [256], [0, 256])
        plt.plot(histogram, color=color, label=f"{color.upper()} Channel")
    plt.title("Color Histogram Analysis")
    plt.xlabel("Pixel Intensity")
    plt.ylabel("Number of Pixels")
    plt.xlim([0, 256])
    plt.legend()
    # Show histogram

```



```
plt.show()
# Wait for key press
cv2.waitKey(0)
cv2.destroyAllWindows()
# Call the function
analyze_histogram("sample.jpg")
```

**OUTPUT:**

